

COURSE SYLLABI

Course Syllabus: FEng 2022 Food analysis and analytical technique

1. **Course Code and Name:** FEng 2022 Food analysis and analytical technique
2. **Credits and Contact hours:** 5CP, 2hrs per week and 3hrs per week (42 hours lecture and 48 hours laboratory)
3. **Instructor's or course coordinator's name:** Mr. Mekonen Tekliye
4. **Text book**
 1. S. Suzanne Nielsen. (2009). Food Analysis, 4th edition: Springer
 2. Nielsen, S. S. (2017). Food analysis, 5th edition: (pp. 3-16): Springer

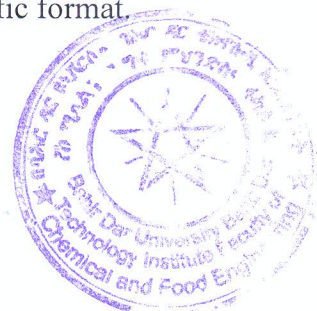
Other supplemental materials

In-class handouts and readings

INSTRUCTIONAL FORMAT: Lectures, assignments, homework, individual work (independent learning), and group works will be given at the end of each chapter

Specific course information

- a. **Catalog description this course covers** This course is designed to introduce the students about the concept of Introduction to food analysis, Trends and demands, Types of samples analyzed, Steps in analysis, Choice and validity of methods, Sampling and sample preparation, Evaluation of analytical data, Proximate analysis of food (theory and practice), Moisture and total solids analysis, Ash, Fat, Protein, Fiber and Carbohydrate analysis. This course also covers the basic analytical tools for determination of proximate analysis in food systems; principles and applications of different instrumental methods for qualitative and quantitative analysis used in the physical, chemical and instrumental examination of food products; selection of the appropriate method for analytes and food systems
 - b. **Prerequisites or co-requisites:**
 - c. **Course Status:** Required /compulsory
5. **Specific goals for the course**
- a. **Specific outcomes of instruction,**
 - Sampling techniques and sample preparation for the analysis of food components
 - The principles, methodology and applications of instrumentation and technology in food analysis
 - Comparison of advanced analytical techniques with conventional techniques in food analysis
 - Application of common techniques for analyzing food components from specific food product to determine proximate composition
 - Assessment of data using statistical analysis and reporting of results in a scientific manner
6. **Student outcomes addressed by the course**
- Apply valid sampling techniques to food materials having widely diverse properties and volumes;
 - Select appropriate analytical techniques for specific food components;
 - Compare advanced and conventional techniques and instruments to analyze chemical and physical properties of foods;
 - Apply a range of chemical analyses of food components;
 - Analyze, interpret and report on results obtained in a scientific format.



7. Brief list of topics to be covered

- Chapter 1:** Introduction to Food Analysis
- Chapter 2:** Evaluation of analytical data
- Chapter 3:** Sampling and Sample Preparation
- Chapter 4:** Compositional Analysis of Foods
- Chapter 5:** Introduction to Instrumental analysis

8. Assessment type, weight and duration

Assessment type		Mark allotted	Duration
Type 1	Group assignment	25%	Before final
Type 2	Mid exam	25%	
Type 3	Final Exam	50%	
Total		100%	

9. Course Policy

All students are expected to abide by the code of conduct of students (article 166 and 166.1.1, of the Senate Legislation of Bahir Dar University May 20, 2005) throughout this course. Academic dishonesty, including cheating, fabrication, and plagiarism will not be tolerated and will be reported to concerned bodies for action.

Class attendance and participation: You are expected to attend class regularly. I will take attendance on regular days during the semester to ensure that students are coming to class, and if you miss class repeatedly, your grade will be affected as it has value. If you miss more than 15% lecture and tutorial and if there is any absence of laboratory class attendance you will not sit for the final exam.

10. Authorization

- a. Prepared instructor's/s' Name: **Mekonen Tekliye**

Signature: _____ Date: _____

- b. Checked course chair's Name: **Mekonen Tekliye**

Signature: _____ Date: _____

- c. Verified chair holder's Name: **Admasu Fenta (PhD)**

Signature: _____ Date: _____

Faculty seal

